



# Upload Files to Dataset

## ! INFO

DataHive supports datasets up to the following classifications:

- RCST: Confidential (Cloud Eligible)
- ISF: Sensitive (Normal)

Datasets on DataHive can be populated via two methods:

- [Method 1: Uploading via the User Interface \(UI\)](#)
- [Method 2: Uploading programmatically via API](#)

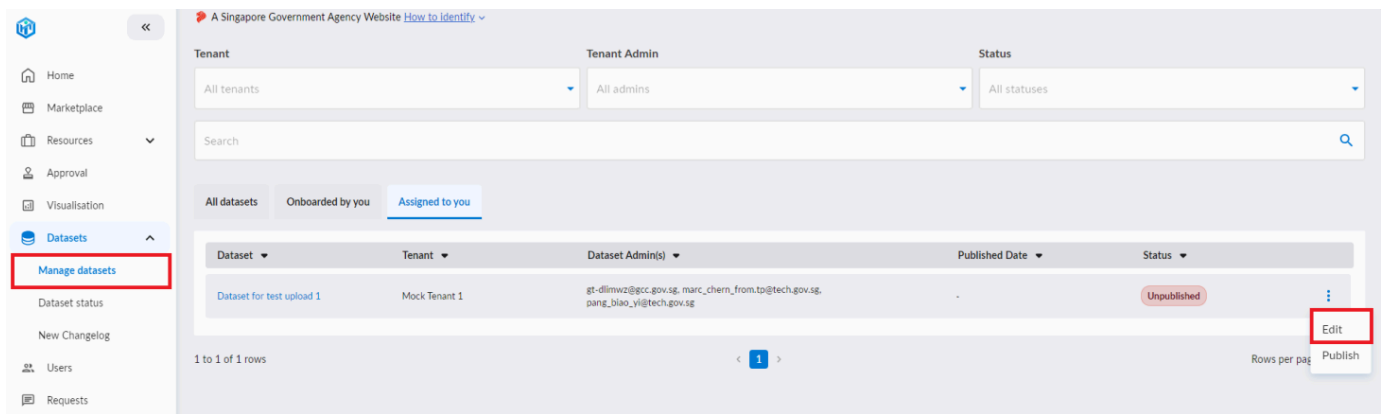
## Method 1: Uploading via the User Interface (UI)

## ! INFO

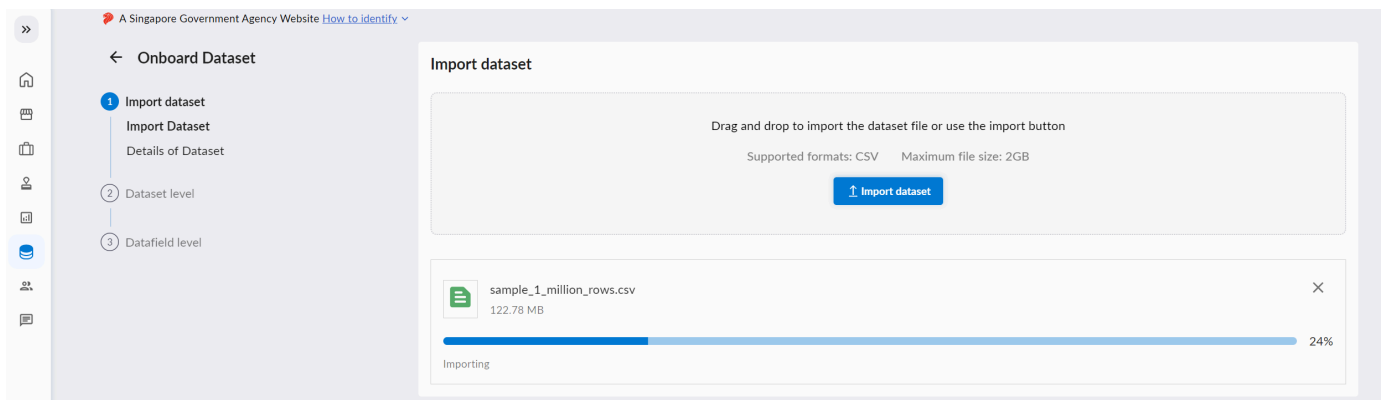
You need to be assigned as **Tenant Admin** in order to upload files to an onboarded dataset. Please note that file upload is **not available for tenants configured to have data shared via Snowflake Private Listing** with DataHive. If your tenant is configured to have data shared via Snowflake Private Listing, refer to [Share Dataset via Snowflake Private Listing](#) for sharing of dataset.

## Accessing the file upload form

To upload files to an onboarded dataset, navigate to "Datasets" > "Manage datasets" and click on the ellipsis associated with the dataset. Select "Edit" from the dropdown menu.



After clicking on the edit button, you will be redirected to the dataset form. You should see the file upload interface as shown below.



Drag and drop your file and a loading bar will appear to notify you of the progress of your upload.

You may upload multiple files but you have to drag and drop them individually (dragging multiple files at once is not supported).

## Method 2: Uploading programmatically via API

### You need to:

- Get a [Tenant Connection](#) to the Tenant that owns the dataset (see [Creating a Service Principal Account](#))
- Get the Tenant Admin to assign the dataset to your Service Principal Account (see [Dataset Assignments](#))

## Implementation Steps

### Step 1: Generate a JWT

Follow the steps [here](#) to generate the JWT token. JWT token will be required for authentication.

## Step 2: Retrieve the Upload Link

Make a GET request to obtain a time-limited upload URL:

```
GET /datasets/{dataset_id}/get_upload_link_spn?file=<name_of_your_file.csv>
```

### Headers

```
Authorization: Bearer {JWT Token}
```

### Response

```
200 OK  
https://api.datahive.gov.sg//...
```



#### **DANGER**

Generate a new upload link for each file upload. Upload links are time-limited and valid for single use only. Reusing a link may result in failed uploads or data corruption.

## Step 3: Upload your file using the Upload Link

Use the SAS URL to upload your `CSV` file via any tool that supports HTTP `PUT` with headers. Common options include:

1. [REST API](#)
2. Azure SDKs (all support SAS-based uploads):
  - [Python](#)
  - [.NET \(C#\)](#)

Alternatively, you can use the sample API script included in your approval email, which automates Steps 1 to 3.

## Using Sample API Script (Optional)

Instead of implementing the three steps above, you can use the pre-built sample script included in your tenant connection approval email. This script automates the entire authentication and upload

process.

 **DANGER**

- Please read and understand the usage of the files as described in this section, especially on handling of `config.json` file.
- Usage of this script in either development or production systems is at the user's discretion.

By extracting the zip file downloaded from the email attachment, you will find 3 files in it.

| File name        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| api_script.py    | This is the main script that you run to call the DataHive API and upload your dataset file.                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| requirements.txt | This file contains the list of python dependencies needed to run the script.                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| config.json      | <p>This file contains the configurations and credentials (user and account IDs) required to run the script to make authenticated calls to DataHive APIs.</p> <p> As this file contains your credentials, please adhere to the following rules when handling this file:</p> <ul style="list-style-type: none"><li>• DO NOT share this file to anyone.</li><li>• DO NOT commit this file to any source code management (SCM) tool such as <code>git</code>.</li></ul> |

To run the script, ensure you have `python` installed and then run the following command to install dependencies:

```
python -m pip install -r requirements.txt
```

Move your private key `rsa_key.p8` to the same folder as the sample script.

Configure the private key passphrase in your environment if your private key is encrypted. If your private key is unencrypted, you can skip this step.

```
export PRIVATE_KEY_PASSPHRASE='<passphrase>'
```

Then run the script with the following arguments:

```
python api_script.py \  
  --private_key_path /path/to/rsa_key.p8 \  
  --dataset_id <your-dataset-id> \  
  --filename /path/to/your_file.csv
```

## Upload Notifications

For each programmatic file upload, DataHive will notify Service Principal Account members and Tenant Admins of the upload outcome via email.

## File format

Files uploaded must adhere to the following format:

1. Only `CSV` files are supported for upload.
2. Each file can be up to `2GB` in size.
3. No line breaks, `\t` (tab) or `\n` (newline) within cells
4. Every file should have a header row with the **same** column names.
5. Every row should have the **same** number of columns as the header row.

The uploaded file(s) will be validated for the above conditions before the upload begins. While the validation is taking place, the progress bar will show a `Validating` status, with the progress set at 0%. This validation may take up to 2 minutes depending on the size of the file. Once completed, the file upload will begin and the progress bar will show the progress of the upload.

The following are examples of valid and invalid CSV files:

Valid CSV file

```
column1,column2,column3  
value1,value2,value3  
value4,value5,value6
```

### Invalid CSV file - Missing header

```
value1,value2,value3  
value4,value5,value6
```

### Invalid CSV file - Row 2 has less columns than header

```
column1,column2,column3  
value1,value2  
value4,value5,value6
```

### Invalid CSV file - Line break/tabs within cell

```
column1,column2,column3  
value1,value\t2,value  
3  
value4,value5,value6
```

## Filling up the Dataset Metadata

After the file upload completes, the Tenant Admin must fill in the necessary information as detailed in [editing and publishing dataset](#) and publish the dataset. Service Principal Accounts can only upload files - they cannot publish datasets.

The headers of the uploaded file will be used to automatically populate the datafields metadata provided that:

1. It is the first file uploaded for the dataset.
2. The datafields metadata has not been filled up before manually or through the [Import Metadata](#) function.

#### WARNING

Tenant Admins should ensure that the ordering, naming and metadata of the datafields entered on the DataHive portal matches the schema in their uploaded file(s).

DataHive does not perform any additional mapping behind the scenes.

Upon publish, the DataHive system would read all the uploaded file(s) and write them to a table in our data warehouse. Users would be able to filter the data based on the publishing date.

The published dataset should be available for subscription on the DataHive portal within half an hour after clicking "Publish", if the publishing job is successful.

Tenant admins publishing datasets will be notified via email when the publish job task has completed (succeeded/failed).

## Adding Data to Published Dataset

After a dataset has been published, you can upload new files using either method described above:

- [Method 1: Tenant Admins can upload via the interface](#)
  - Clicking on "Save" with newly uploaded file(s) would trigger a publish job to append the data from these new files to the existing table previously created at publish time.
  - Tenant Admins do not need to click "Publish" again once they have published a dataset. A publish job is automatically ran if there are new files uploaded to the DataHive platform.
- [Method 2: Service Principal Accounts can upload programmatically](#)
  - Processing starts automatically after successful upload.

The following conditions currently apply after a dataset has been published:

1. Uploaded files cannot be deleted.
2. Datafields cannot be removed or reordered.

If you need to apply any of the above mentioned changes to a dataset after it has been published, please refer to the guide on [Un-publishing Dataset](#).

## Additional Information

For a comprehensive overview of the entire dataset onboarding process via file uploads, please refer to the steps and frequently asked questions available [here](#).